# ITSimplera Institute

## **WEEK 07**: Enterprise VPN, Firewall, and Secure Access Implementation

**Name**: Babar Khan

**Reg No:** NETB01-2275

**Dated:** 15-08-2026

# Table of Contents

# Enterprise VPN, Firewall, and Secure Access Implementation

## 1. Introduction

This report documents the design, implementation, and verification of a multi-site enterprise network lab built to demonstrate secure inter-site connectivity. The topology models a small enterprise with a Head Office (HQ) and a remote branch (BRANCH-1) that connect to each other only through a simulated Internet Service Provider (MAIN-ROUTER), reflecting how real WAN edges typically interconnect.

The core objective of the lab was to secure that inter-site connectivity end-to-end: routing between sites is established and protected with OSPF, the link between HQ and BRANCH-1 is encrypted with a site-to-site IPsec VPN, traffic entering and leaving each site is controlled by a zone-based firewall, and device management access is hardened and restricted to SSH with centralized authentication and logging. Together these layers implement defense-in-depth around the enterprise WAN edge, rather than relying on any single control.

The sections below define the key technologies involved, present the lab topology and device inventory, provide the IP addressing plan, walk through verification evidence for each implemented feature, summarize the troubleshooting encountered, and close with a conclusion on the outcome of the exercise.

## 2. Key Concepts and Definitions

**Virtual Private Network (VPN):** A method of extending a private network across a public or untrusted network (such as the Internet) by creating an encrypted tunnel between two endpoints. In this lab, a site-to-site IPsec VPN is established between the HQ and BRANCH-1 routers using IKE/ISAKMP for key negotiation and an ESP transform set for encrypting and authenticating traffic between the 192.168.10.0/24 and 192.168.20.0/24 LANs.

**Firewall:** A network security control that inspects and enforces policy on traffic crossing a defined boundary. This lab uses Cisco's Zone-Based Policy Firewall (ZBFW), where interfaces are placed into security zones (INSIDE, OUTSIDE), and class-maps/policy-maps applied to zone-pairs decide which traffic is inspected, permitted, or dropped between those zones.

**OSPF Authentication:** A routing-security feature of Open Shortest Path First (OSPF) that requires neighboring routers to validate each other's identity in this lab using MD5 message-digest authentication before they will form an adjacency and exchange routes. This prevents rogue or unauthorized routers from injecting false routing information into the network.

**Monitoring:** The ongoing collection and review of operational and security data (interface states, neighbor relationships, VPN/firewall session counters, authentication attempts) to confirm that a network is behaving as designed and to detect problems quickly. In this lab, monitoring is performed through router show commands (OSPF, ISAKMP, IPsec, zone-pair sessions) alongside centralized log collection.

**Syslog:** A standard protocol for forwarding log messages from network devices to a centralized log server. In this lab, the routers are configured to send their log messages to a syslog server hosted on the network-toolkit Docker host, giving a single, centralized record of system and security events across all devices.

## 3. Network Topology

The lab consists of three Cisco c3725 routers, two Cisco IOU switches, and a set of end devices and Docker-hosted services. MAIN-ROUTER represents the ISP/transit core; HQ and BRANCH-1 represent the two enterprise sites, each with its own LAN, connecting to MAIN-ROUTER over a dedicated /30 transit link (no direct HQ–BRANCH-1 link exists all inter-site traffic transits MAIN-ROUTER).

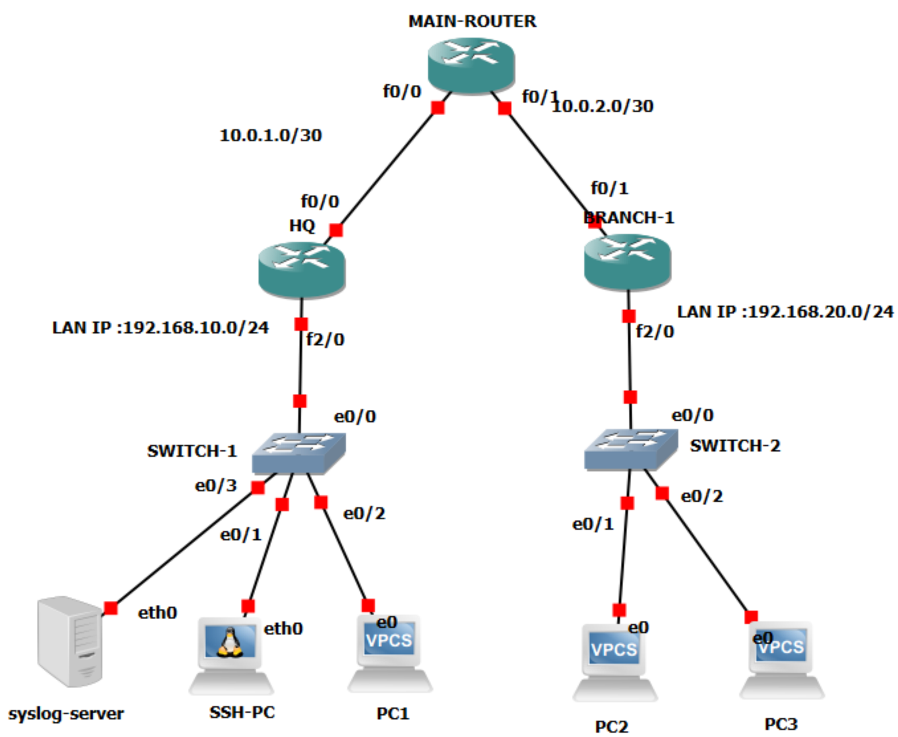


*Figure 1 – Lab topology: MAIN-ROUTER, HQ, and BRANCH-1 (Cisco c3725), SWITCH-1/SWITCH-2 (Cisco IOU), end hosts, and the syslog server.*

### 3.1 Devices Used

| Device | Quantity | Function in the Lab |
|---|---|---|
| Cisco c3725 Router | 3 (MAIN-ROUTER, HQ, BRANCH-1) | Provide inter-site routing (OSPF), IPsec VPN termination, and the zone-based firewall |
| Cisco IOU Switch | 2 (SWITCH-1, SWITCH-2) | Layer-2 access switching for the HQ and BRANCH-1 LANs |
| AAA Docker container | 1 | Centralized authentication, authorization, and accounting (AAA) for device management |
| Network-toolkit Docker server | 1 | Hosts monitoring/logging tools, including the syslog collector used for verification |

## 4. IP Allocation Table

The addressing plan below reflects the confirmed interface and host addressing used across the topology. LAN hosts drawn from a subnet's address pool are noted where an individual host address was not directly confirmed in verification output.

| Device | Interface | IP Address | Network / Purpose | Role |
|---|---|---|---|---|
| MAIN-ROUTER (c3725) | f0/0 | 10.0.1.1/30 | MAIN ↔ HQ transit link | ISP / Core |
| MAIN-ROUTER (c3725) | f0/1 | 10.0.2.1/30 | MAIN ↔ BRANCH-1 transit link | ISP / Core |
| HQ (c3725) | f0/0 | 10.0.1.2/30 | HQ ↔ MAIN transit link | Site router |
| HQ (c3725) | f2/0 | 192.168.10.1/24 | HQ LAN gateway | Site router |
| BRANCH-1 (c3725) | f0/1 | 10.0.2.2/30 | BRANCH-1 ↔ MAIN transit link | Site router |
| BRANCH-1 (c3725) | f2/0 | 192.168.20.1/24 | BRANCH-1 LAN gateway | Site router |
| SWITCH-1 (Cisco IOU) | e0/0–e0/3 | 192.168.10.0/24 (L2) | HQ access switch | Access layer |
| SWITCH-2 (Cisco IOU) | e0/0–e0/2 | 192.168.20.0/24 (L2) | BRANCH-1 access switch | Access layer |
| PC1 (VPCS) | eth0 | 192.168.10.11/24 | HQ LAN test host | End device |
| SSH-PC (Linux host) | eth0 | 192.168.10.0/24 pool | HQ management / SSH test client | End device |
| syslog-server | eth0 | 192.168.10.0/24 pool | Centralized logging target | Monitoring |
| PC2 (VPCS) | eth0 | 192.168.20.10/24 | BRANCH-1 LAN test host | End device |
| PC3 (VPCS) | eth0 | 192.168.20.0/24 pool | BRANCH-1 LAN test host | End device |
| AAA-1 / Docker (AAA + toolkit) | docker0 | Docker bridge network | TACACS+/RADIUS AAA and network-toolkit services | Management host |

# 5. Implementation and Verification

This section presents command-line verification evidence for each security control implemented in the lab, organized by feature.

## 5.1 OSPF Routing and Routing Security

OSPF is enabled in Area 0 across all three routers with MD5 authentication configured on the inter-router links, and passive-interface behavior verified on the LAN-facing interfaces so that OSPF hellos are not advertised toward end hosts. Router IDs were confirmed as 1.1.1.1 (MAIN-ROUTER), 2.2.2.2 (HQ), and 3.3.3.3 (BRANCH-1), with full adjacencies formed over the transit links.

```
MAIN-ROUTER#show ip ospf interface f0/0
FastEthernet0/0 is up, line protocol is up
  Internet Address 10.0.1.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State WAITING, Priority 1
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:06
    Wait time before Designated router selection 00:00:06
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 0
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 1
MAIN-ROUTER#show ip ospf neighbor
```

*Figure 2 – MAIN-ROUTER: OSPF interface state on f0/0 with MD5 authentication enabled.*

```
HQ#show ip ospf interface f0/0
FastEthernet0/0 is up, line protocol is up
  Internet Address 10.0.1.2/30, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 2.2.2.2, Interface address 10.0.1.2
  Backup Designated router (ID) 1.1.1.1, Interface address 10.0.1.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:02
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 1
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.1  (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 1
HQ#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:36    10.0.1.1        FastEthernet0/0
HQ#
```

*Figure 3 – HQ: OSPF interface/neighbor verification – full adjacency with MAIN-ROUTER (1.1.1.1).*

```
BRANCH-1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:38    10.0.2.1        FastEthernet0/1
BRANCH-1#show ip ospf interface f0/0
%OSPF: OSPF not enabled on FastEthernet0/0
BRANCH-1#show ip ospf interface f0/1
FastEthernet0/1 is up, line protocol is up
  Internet Address 10.0.2.2/30, Area 0
  Process ID 1, Router ID 3.3.3.3, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 3.3.3.3, Interface address 10.0.2.2
  Backup Designated router (ID) 1.1.1.1, Interface address 10.0.2.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:00
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 2
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.1  (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 1
BRANCH-1#
```

*Figure 4 – BRANCH-1: OSPF neighbor table and interface state – full adjacency with MAIN-ROUTER, confirming f0/0 has no OSPF neighbors (LAN-facing, passive).*

## 5.2 Site-to-Site IPsec VPN

A site-to-site IPsec VPN protects traffic between the HQ and BRANCH-1 LANs across the MAIN-ROUTER transit path. ISAKMP Phase 1 negotiates in QM_IDLE/ACTIVE state between the HQ (10.0.1.2) and

BRANCH-1 (10.0.2.2) peers, and Phase 2 IPsec SAs show active encapsulation/decapsulation of traffic between 192.168.10.0/24 and 192.168.20.0/24 using esp-256-aes with esp-sha-hmac.

```
HQ#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
10.0.2.2        10.0.1.2        QM_IDLE           1001    0 ACTIVE

IPv6 Crypto ISAKMP SA

HQ#
```

*Figure 5 – HQ: ISAKMP SA in QM_IDLE / ACTIVE state with BRANCH-1 (10.0.2.2).*

```
HQ#show crypto ipsec sa

interface: FastEthernet0/0
    Crypto map tag: VPN-MAP, local addr 10.0.1.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
   remote ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
   current_peer 10.0.2.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 4, #pkts encrypt: 4, #pkts digest: 4
    #pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 3
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0

     local crypto endpt.: 10.0.1.2, remote crypto endpt.: 10.0.2.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0x98312370(2553357168)

     inbound esp sas:
      spi: 0xF671F80F(4134664207)
        transform: esp-256-aes esp-sha-hmac ,
 --More--
```

*Figure 6 – HQ: IPsec SA detail – active encaps/decaps counters and esp-256-aes / esp-sha-hmac transform.*

```
BRANCH-1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
10.0.2.2        10.0.1.2        QM_IDLE           1001    0 ACTIVE

IPv6 Crypto ISAKMP SA
```

*Figure 7 – BRANCH-1: ISAKMP SA in QM_IDLE / ACTIVE state with HQ (10.0.1.2).*

```
BRANCH-1#show crypto ipsec sa

interface: FastEthernet0/1
    Crypto map tag: VPN-MAP, local addr 10.0.2.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
   remote ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
   current_peer 10.0.1.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 10, #pkts encrypt: 10, #pkts digest: 10
    #pkts decaps: 12, #pkts decrypt: 12, #pkts verify: 12
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 10.0.2.2, remote crypto endpt.: 10.0.1.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/1
     current outbound spi: 0xF671F80F(4134664207)

     inbound esp sas:
      spi: 0x98312370(2553357168)
        transform: esp-256-aes esp-sha-hmac ,
```

*Figure 8 – BRANCH-1: IPsec SA detail – matching encaps/decaps counters and transform set.*

```
MAIN-ROUTER#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status

IPv6 Crypto ISAKMP SA
```

*Figure 9 – MAIN-ROUTER: empty ISAKMP SA table, confirming the VPN tunnel terminates only at HQ and BRANCH-1 the ISP router simply forwards encrypted transit traffic.*

## 5.3 Zone-Based Firewall

Each site router implements a Zone-Based Policy Firewall with an INSIDE zone (LAN-facing interface) and an OUTSIDE zone (WAN-facing interface toward MAIN-ROUTER). Zone-pairs IN-TO-OUT and OUT-TO-IN apply an inspect policy-map that classifies and inspects TCP, UDP, and ICMP traffic, with a default class-map dropping any unmatched traffic.

**HQ**

```
HQ#show zone security
zone self
  Description: System defined zone


zone INSIDE
  Member Interfaces:
    FastEthernet2/0


zone OUTSIDE
  Member Interfaces:
    FastEthernet0/0


HQ#
```

*Figure 10 – HQ: zone security configuration – INSIDE (Fa2/0) and OUTSIDE (Fa0/0).*

```
HQ#show zone-pair security
Zone-pair name IN-TO-OUT
    Source-Zone INSIDE  Destination-Zone OUTSIDE
    service-policy INSIDE-TO-OUTSIDE-POLICY
Zone-pair name OUT-TO-IN
    Source-Zone OUTSIDE  Destination-Zone INSIDE
    service-policy INSIDE-TO-OUTSIDE-POLICY

HQ#
```

*Figure 11 – HQ: zone-pairs IN-TO-OUT and OUT-TO-IN bound to the INSIDE-TO-OUTSIDE-POLICY service policy.*

```
        0 packets, 0 bytes
HQ#show policy-map type inspect zone-pair sessions
 Zone-pair: IN-TO-OUT

  Service-policy inspect : INSIDE-TO-OUTSIDE-POLICY

    Class-map: INSIDE-TO-OUTSIDE-TRAFFIC (match-any)
      Match: protocol tcp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol udp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol icmp
        2 packets, 128 bytes
        30 second rate 0 bps
      Inspect

    Class-map: class-default (match-any)
      Match: any
      Drop
        0 packets, 0 bytes
 Zone-pair: OUT-TO-IN

  Service-policy inspect : INSIDE-TO-OUTSIDE-POLICY

    Class-map: INSIDE-TO-OUTSIDE-TRAFFIC (match-any)
      Match: protocol tcp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol udp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol icmp
        0 packets, 0 bytes
```

*Figure 12 – HQ: inspect policy-map session counters for both zone-pairs, showing matched ICMP traffic and the class-default drop rule.*

## BRANCH-1

```
BRANCH-1#show zone security
zone self
  Description: System defined zone


zone INSIDE
  Member Interfaces:
    FastEthernet2/0


zone OUTSIDE
  Member Interfaces:
    FastEthernet0/1
```

*Figure 13 – BRANCH-1: zone security configuration – INSIDE (Fa2/0) and OUTSIDE (Fa0/1).*

```
BRANCH-1#show zone-pair security
Zone-pair name IN-TO-OUT
    Source-Zone INSIDE  Destination-Zone OUTSIDE
    service-policy INSIDE-TO-OUTSIDE-POLICY
Zone-pair name OUT-TO-IN
    Source-Zone OUTSIDE  Destination-Zone INSIDE
    service-policy INSIDE-TO-OUTSIDE-POLICY
```

*Figure 14 – BRANCH-1: zone-pairs IN-TO-OUT and OUT-TO-IN bound to the INSIDE-TO-OUTSIDE-POLICY service policy.*

```
 Service-policy inspect : INSIDE-TO-OUTSIDE-POLICY

   Class-map: INSIDE-TO-OUTSIDE-TRAFFIC (match-any)
     Match: protocol udp
       0 packets, 0 bytes
       30 second rate 0 bps
     Match: protocol tcp
       0 packets, 0 bytes
       30 second rate 0 bps
     Match: protocol icmp
       0 packets, 0 bytes
       30 second rate 0 bps
     Inspect

   Class-map: class-default (match-any)
     Match: any
     Drop
       0 packets, 0 bytes
Zone-pair: OUT-TO-IN

 Service-policy inspect : INSIDE-TO-OUTSIDE-POLICY

   Class-map: INSIDE-TO-OUTSIDE-TRAFFIC (match-any)
     Match: protocol udp
       0 packets, 0 bytes
       30 second rate 0 bps
     Match: protocol tcp
       0 packets, 0 bytes
       30 second rate 0 bps
     Match: protocol icmp
       2 packets, 128 bytes
       30 second rate 0 bps
     Inspect

   Class-map: class-default (match-any)
     Match: any
     Drop
       0 packets, 0 bytes
BRANCH-1#
```

*Figure 15 – BRANCH-1: inspect policy-map session counters confirming the firewall policy is active on both zone-pairs.*

### 5.4 Secure Device Management and Hardening

Device management access was hardened by disabling unnecessary/legacy services (including Telnet) and restricting remote administration to SSH only, backed by centralized AAA. The verification below confirms Telnet is refused while SSH remains reachable.

```
Connection to 192.168.10.1 closed by remote host.
Connection to 192.168.10.1 closed.
root@SSH-PC:~# curl http://192.168.10.1
curl: (7) Failed to connect to 192.168.10.1 port 80: Connection refused
root@SSH-PC:~#
```

*Figure 16 – Confirmation that unnecessary services have been disabled on the router.*

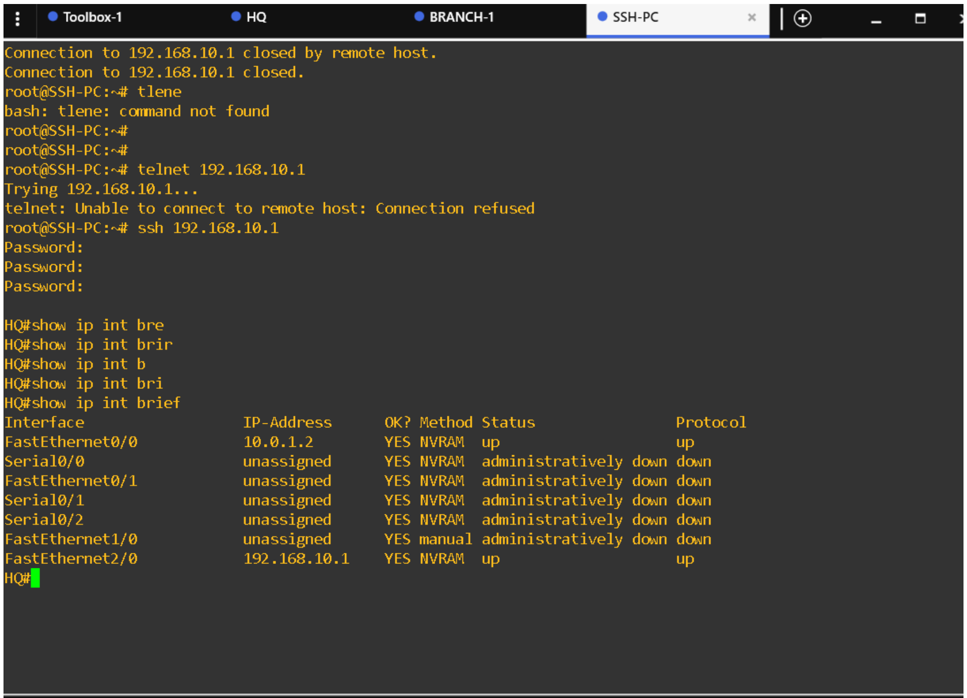


*Figure 17 – SSH-PC: Telnet to the HQ router (192.168.10.1) is refused, while SSH successfully reaches the login prompt confirming hardening is effective and secure management access works as intended.*

## 5.5 End-to-End Connectivity

With OSPF routing, the IPsec VPN, and the zone-based firewall all active together, end-to-end reachability between the HQ and BRANCH-1 LANs was confirmed in both directions.

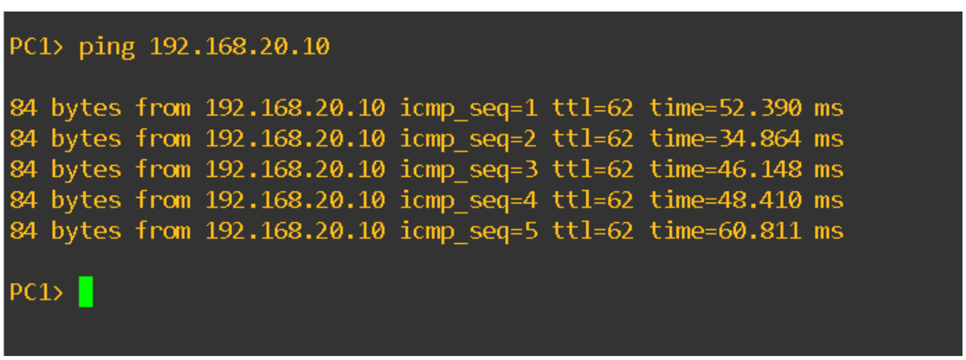


*Figure 18 – PC1 (HQ LAN) successfully pinging PC2 (BRANCH-1 LAN, 192.168.20.10) across the encrypted, firewalled inter-site path.*

```
PC2> ping 192.168.10.11

192.168.10.11 icmp_seq=1 timeout
192.168.10.11 icmp_seq=2 timeout
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=48.376 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=54.109 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=47.745 ms

PC2>
```

*Figure 19 – PC2 (BRANCH-1 LAN) pinging back to the HQ LAN (192.168.10.11); the first two probes time out while the firewall/VPN session establishes, then succeed expected first-packet behavior for stateful inspection and IKE negotiation.*

# 6. Troubleshooting Report

A number of practical challenges came up during implementation and testing. The table below summarizes the most significant ones and how each was resolved.

| Challenge | Symptom / Root Cause | Resolution |
|---|---|---|
| Zone-based firewall policy design | Traffic between the INSIDE and OUTSIDE zones is dropped by default until an explicit zone-pair and inspect policy exist in the direction of the traffic; a policy applied to only one direction leaves the return path (or the opposite initiation direction) unprotected/blocked. | Defined INSIDE and OUTSIDE zones on the LAN and WAN-facing interfaces, then built matching class-maps/policy-maps and bound them to both the IN-TO-OUT and OUT-TO-IN zone-pairs, so traffic is correctly inspected in both directions. Verified with show zone security, show zone-pair security, and show policy-map type inspect zone-pair sessions on HQ and BRANCH-1. |
| SSH access from the test PC | The Linux-based SSH-PC did not have the required client packages available out of the box, so initial connection attempts failed at the shell level before any network issue could even be tested. | Installed the missing SSH client packages on SSH-PC, then re-tested. Telnet to the routers correctly returned "Connection refused" (confirming the hardening step of disabling Telnet), while SSH successfully reached the router login prompt, confirming secure management access was working as designed. |
| Legacy IOS crypto support | The c3725 IOS image does not support SHA-256 hashing or Diffie-Hellman group 14 for ISAKMP, and modern SSH clients default to key exchange/host-key | Used hash SHA with DH group 5 and esp-sha-hmac for the IPsec transform set, and added legacy KexAlgorithms/HostKeyAlgorithms/Ciphers |

| | | |
|---|---|---|
| | algorithms the old IOS SSH stack does not offer. | overrides to the SSH client configuration on the Docker management host so it could negotiate with the routers. |

# 7. Conclusion

This lab successfully implemented and verified a layered set of enterprise security controls across a multi-site topology built from Cisco c3725 routers, Cisco IOU switches, and Docker-hosted AAA/monitoring services. OSPF routing with MD5 authentication and passive interfaces provided a secure and predictable routing base; a site-to-site IPsec VPN encrypted all traffic between the HQ and BRANCH-1 LANs across the shared transit path; a zone-based firewall enforced explicit, inspected policy at each site's network edge; and device management was hardened to SSH-only access with centralized logging to a syslog server.

Verification evidence OSPF neighbor and interface state, ISAKMP/IPsec security associations, zone-pair and policy-map session counters, Telnet-refused/SSH-accepted management access, and successful bidirectional ping tests between the two LANs confirms that all implemented features are functioning together as an integrated, defense-in-depth design rather than as isolated configurations. The troubleshooting encountered, particularly around bidirectional zone-firewall policy and client-side SSH tooling, reinforced the importance of testing each control from both directions and at every layer of the stack. Remaining work noted for a future phase includes NTP time synchronization and SNMPv3 monitoring, which were intentionally scoped out of this task.